

C2_06 Gate A Protected Review - Astra RC4

Logistics Shipments. Open only after a saved genuine Gate A attempt.

1. Objective fixture controls

Control	Expected	Why it matters
Fact grains	100 shipments; 200 events = exactly 2 events/shipment	Shipment and event are separate processes/grains.
Join trap	Shipment charge 113,006.49; naive event join sum 226,012.98	Joining event rows then summing shipment amount doubles the measure.
Unknown FK	H99 on 1 shipment; charge 1,795.87	Unknown hub must be surfaced/resolved or mapped to a controlled unknown member.
Target fact	21 unique Hub-Month rows; total target 3,255	Target is higher-grain and must not repeat on shipment/event rows.
History	H03 Metro -> Regional effective 2026-05-15	Type 2 versioning must attribute facts by book_date.
H03 evidence	8 shipments before change / 7 after	Historical classification cannot be reconstructed from only the current master.
Status	50 Delayed / 50 Delivered	Useful independent fact control.

2. Expected reasoning path

Model Shipment and ShipmentEvent as separate facts (or preserve event grain explicitly); do not aggregate shipment measures after a one-to-many event join.

Use Hub as a dimension; target remains Hub-Month fact. H99 requires explicit unknown/FK handling.

For H03, close Metro version and open Regional version on the effective boundary; facts before/after use the appropriate surrogate key.

Choose staging/curated/presentation and warehouse/lakehouse/batch choices from replay, governance and latency needs; document lineage to KPI.

3. Critical failures

- One combined fact mixes shipment and event grains.
- Charge doubles after event join.
- Hub-Month target repeats per shipment/event.
- H03 current value overwrites history.
- SCD2 intervals overlap or two rows are current.
- H99 is silently dropped.
- Pipeline publishes despite unresolved FK/reconciliation blocker.

4. Competency floors

Competency	Required proof
Process / grain	Facts, dimensions and targets represent declared processes/grains; no fan-out or repeated target.
Keys / SCD history	Business/surrogate keys, unknown-member handling and non-overlapping historical intervals are correct.
Layer / store / latency	ETL/ELT, replay layers, platform and batch/stream choice follow requirements and recovery needs.
Lineage / metadata	Source-to-KPI path, owner/grain/schema/rule metadata are sufficient to explain the output.
Quality / release	Reconciliations and integrity checks have severity/owner/action; blocking failure stops publication.

Changed repair proof

This protected review opens only after a saved genuine Gate A attempt. Close it and complete a changed fresh retry on new logistics history/target/unknown-key evidence. Save retry evidence. Gate B is a different E-commerce Returns reassessment and opens only when explicitly assigned after repair.